Fibre bundle analysis of topological charges in spontaneously broken gauge theories

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We study the topological properties of spontaneously broken gauge theories in the context of fibre bundle theory. In particular, we discuss the conditions under which the topological charges of gauge and Higgs fields are the same.

I. INTRODUCTION

The aim of the present paper is to study topological properties of spontaneously broken gauge theories, giving an explicit geometrical description in terms of connections and cross sections in a principal and associated fibre bundles.

As it was noted by Popov,[1] this approach not only enables one to "calculate" but renders calculations more transparent.

In Sec. II we study the simple case of an Abelian Higgs theory in $d=2$ (Euclidean) dimensions and show how the geometrical interpretation leads naturally to the identification of the topological charges of both gauge fields and Higgs field.

The discussion of the general case is done in Sec. III: for a compact Lie group G, conditions under which a single K-uple of integers labels the topological charges of gauge and Higgs fields are obtained. These conditions, already obtained by Woo[2] for the case $d=4$ (Euclidean) dimensions, using rather different techniques, arises in a transparent manner in the context of the fibre bundle theory. Several examples are discussed at the end of this section.

II. THE ABELIAN CASE

We will consider in this section the Abelian Higgs model: an Abelian gauge field A_μ coupled to a complex scalar field ϕ, with dynamics determined by the Lagrangian density

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + (\partial_\mu + iA_\mu)\phi^*(\partial_\mu - iA_\mu)\phi - \mathcal{U}(\phi), \quad (2.1)$$

where

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad \phi = \phi_1 + i\phi_2, \quad (2.2)$$

and

$$\mathcal{U}(\phi) = \tfrac{1}{2}\lambda(\phi^*\phi - \mu^2/2\lambda)^2. \quad (2.3)$$

This Lagrangian density is invariant under the local gauge transformations generated by the group $G=\mathcal{U}(1)$; we are interested in the case $\mu^2>0$, that is, when the symmetry is spontaneously broken. It was noted by Nielsen and Olesen[3] that this model allows for vortex solutions-static, cylindrically symmetric field configurations of finite energy per unit length. The Nielsen—Olesen solution depends only on two variables. Then, the vortex field can be considered as a pseudoparticle configuration (with finite action) in the Euclidean version of the two-dimensional theory. If the action is to be finite, then, at Euclidean infinity (that is, on the sphere S^1_∞)

$$\lim_{r\to\infty} F_{\mu\nu} = 0, \quad (2.4)$$

$$\lim_{r\to\infty} (\partial_\mu - iA_\mu)\phi = 0, \quad (2.5)$$

$$\lim_{r\to\infty} \mathcal{U}(\phi) = 0, \quad (2.6)$$

where $r^2 = x_1^2 + x_2^2$.

These conditions can be interpreted geometrically in terms of the theory of fibre bundles. To this end, we consider S^1_∞ as the base space of a principal fibre bundle $P(S^1_\infty, \mathcal{U}(1))$ with structure group $\mathcal{U}(1)$. If we call $\mathcal{J}$ the associated vector bundle with fibre $\mathbb{C}$—the group $\mathcal{U}(1)$ acts on the left on $\mathbb{C}$ by usual multiplication—then the scalar field ϕ can be considered as a global cross section on $\mathcal{J}$[4].

Because $\phi \neq 0$ on S^1_∞ [from Eq. (2.3) and condition (2.6)] $\mathcal{J}$ is trivial and so is P.

The one form $A_\mu dx^\mu$ defines a connection on P. We can then consider on $\mathcal{J}$ the covariant derivative D induced by the connection [its explicit expression is $(\partial_\mu - iA_\mu)$].

Let γ be a given curve in S^1_∞ from x_0 to x. If $(x_0, g) \in P_{x_0}$ and $(x_0, v) \in \mathcal{J}_{x_0}$ [$\mathcal{J}_{x_0}(P_{x_0})$ is the fibre of $\mathcal{J}(P)$ over x_0], we will call $(x, T_\gamma g) \in P_x$ the parallel displacement of (x_0, g) along γ, given by the connection $A_\mu dx^\mu$, and $(x, \widetilde{T}_\gamma v) \in \mathcal{J}_x$ the parallel displacement of (x, v) along γ given by D. Since D is induced by the connection,

$$\widetilde{T}_\gamma v = (T_\gamma g)v/g \quad (2.7)$$

and

Lemma: for any curve γ' in S^1_∞ from x_0 to x, $T'_\gamma g = T_\gamma g \;\forall\; g \in \mathcal{U}(1)$ and $\widetilde{T}'_\gamma v = \widetilde{T}_\gamma v \;\forall\; v \in \mathbb{C}$.

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Proof: From Eq. (2.5), $\phi(x)=\tilde{T}_\gamma\phi(x_0)$; then for any $g\in U(1)$

$$\phi(x)=(T_\gamma g)\phi(x_0)/g. \tag{2.8}$$

But Eq. (2.5) also implies that $\phi(x)=\tilde{T}'_\gamma\phi(x_0)$. Hence, $\phi(x)=(T'_\gamma g)\,\phi(x_0)/g$. Since $\phi(x_0)\neq 0$, it follows $T_\gamma g=T'_\gamma g$. Now, from (2.7) it is evident that $\tilde{T}_\gamma v=\tilde{T}'_\gamma v$. That is, the parallel displacement from x_0 to x is independent of the particular choice of γ in S^1_∞.

Hence, if we fix a point $x_0\in S^1_\infty$, the given connection $A_\mu dx^\mu$ can be identified with a well-defined mapping

$$\tilde{A}:S^1_\infty\rightarrow U(1),$$
$$\tilde{A}(x)=T_\gamma(I), \tag{2.9}$$

where I is the identity of $U(1)$ and γ is any curve from x_0 to x in S^1_∞. We note that the explicit form of $\tilde{A}(x)$ reads

$$\tilde{A}(x)=\exp(i\int_{x_0}^x A_\mu dx^\mu).$$

Taking $g=I$ in Eq. (2.7), we have

$$\phi(x)=\tilde{A}(x)\phi(x_0). \tag{2.10}$$

Now, condition (2.6) shows that ϕ defines a mapping $\phi:S^1_\infty\rightarrow S^1$. Because $U(1)\equiv S^1$, $\tilde{A}$ can be also considered as a mapping $\tilde{A}:S^1_\infty\rightarrow S^1$. Then, relation (2.10) implies that ϕ and $\tilde{A}$ *belong to the same class in* $\Pi(S^1)\equiv Z$. Hence, *the same integer characterizes topologically the gauge field A_μ and the scalar field ϕ.*

This integer, the topological charge "n" is related to the quantization of the magnetic flux of the vortex. In terms of the gauge field A_μ it is given by the expression

$$n=\frac{1}{2\pi}\oint_{S^1_\infty}A_\mu dx^\mu$$

that, according to Stokes theorem, can be written as

$$n=\frac{1}{4\pi}\int_{\mathbb{R}^2}F_{\mu\nu}dx^\mu\wedge dx^\nu.$$

Our derivation of the topological equivalence between Higgs and gauge fields can be understood intuitively as follows: The principal fibre bundle P can be visualized as a torus (A) (see Fig. 1). Then the topological charge, associated with the gauge field counts the number of times that the curve $\bar{\gamma}$ joining the points $(x,\tilde{A}(x))$ in P winds around the torus. On the other hand, we can take on each fibre $\mathbb{C}$ of the vector bundle $\mathcal{F}$ the circle $|z|=a_0(a_0=(\mu^2/2\lambda)^{1/2})$, that is, all the possible values of ϕ on S^1_∞. We thus obtain another torus (B) and we can take on it the curve $\tilde{\tau}$ joining the points $(x,\phi(x))$. The topological charge associated with the Higgs field corresponds to the number of times the curve $\tilde{\tau}$ winds around this torus. Relation (2.10) states the equality between both numbers.

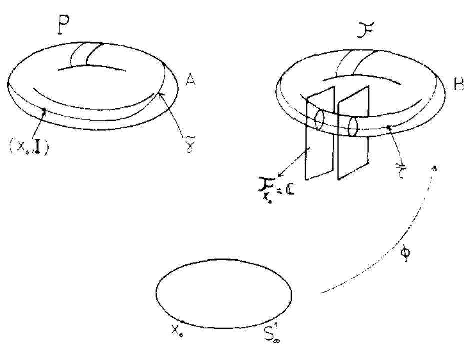

FIG. 1. We represent at the bottom the base space S^1_∞ of the principal fiber bundle P (torus A). Taking on each fibre C of the vector bundle $\mathcal{F}$ the circles $|Z|=a_0$, we obtain another torus (B).

III. HIGHER-DIMENSIONAL CASES

In this section, we consider the general case of a spontaneously broken gauge theory in a Euclidean d-dimensional space $(d>2)$. Let G be a compact Lie group and

$$L=-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}-\tfrac{1}{2}D_\mu\phi D^\mu\phi-U(\phi) \tag{3.1}$$

a Yang-Mills Lagrangian density for G. The field ϕ transforms in accordance with an N-dimensional unitary irreducible representation of G, and the fields $A_\mu=(A_\mu^1,\ldots,A_\mu^n)$ assume values in the algebra of the Lie Group G. $F_{\mu\nu}=\partial_\mu A_\nu-\partial_\nu A_\mu-i[A_\mu,A_\nu]$, D_μ is the covariant derivative, and $U(\phi)$ is a G-invariant function which has a minimum (this minimum is assumed to be equal to zero). As in the Abelian case, finiteness of the action imposes the following conditions on S^{d-1}_∞:

$$\lim_{r\to\infty}F_{\mu\nu}=0, \tag{3.2}$$
$$\lim_{r\to\infty}D_\mu\phi=0, \tag{3.3}$$
$$\lim_{r\to\infty}U(\phi)=0, \tag{3.4}$$

where $r^2=x_1^2+\ldots+x_d^2$. This situation corresponds to the following geometrical description: a flat connection $A_\mu dx^\mu$ [condition (3.2)] on a principal fibre bundle $P(S^{d-1}_\infty,G)$ with base manifold S^{d-1}_∞ and structure group G.

Since S^{d-1}_∞ is simply connected $(d>2)$ and the connection is flat, then it follows that P is trivial and also that the parallel displacement along a curve γ in S^{d-1}_∞ from x_0 to x, induced by the connection is independent of the particular choice of the curve γ. (These results follow from the general theory of flat connections, see, for example, Ref. 4).

As we did in Sec. II, let us fix a point $x_0\in S^{d-1}_\infty$. The connection $A_\mu dx^\mu$ determines a well defined mapping

$$\tilde{A}:S^{d-1}_\infty\rightarrow G, \tag{3.5}$$
$$\tilde{A}(x)=T_\gamma I, \tag{3.6}$$

where I is the identity of G and $(x,T_\gamma I)\in P_x$ is the parallel displacement of $(x_0,I)\in P_{x_0}$ along any curve γ in S^{d-1}_∞ from x_0 to x.

Let $\mathcal{F}$ be the associated bundle with fibre K, where K is a vector space-tensor space—according to the choice of ϕ. (The group G acts on the left on K in the usual way).

Let us call D the covariant derivative induced on $\mathcal{F}$ by the connection given on P. From Eq. (3.3) the Higgs field ϕ can be considered as a global cross section of the bundle $\mathcal{F}$ with

$$\phi(x)=\tilde{T}_\gamma \phi(x_0), \tag{3.7}$$

where $(x, \tilde{T}_\gamma\phi(x_0)) \in \mathcal{J}_x$ is the parallel displacement of $(x_0, \phi(x_0)) \in \mathcal{J}_{x_0}$ along any curve γ from x_0 to x, induced by D.

Now Eq. (3.7) implies that

$$\phi(x)=\tilde{A}(x)\phi(x_0). \tag{3.8}$$

Let us call $a_0 \in K$ a fixed element of K satisfying $\mathcal{U}(a_0)=0$ on S_∞^{d-1}. Because $\mathcal{U}(\phi)$ is G-invariant, ga_0 also satisfies condition (3.4) for every $g\in G$. We will assume that all the zeros of $\mathcal{U}$ on S_∞^{d-1} are of this form.[5] Hence, the set of zeros of $\mathcal{U}$ can be identified with the left coset space G/H where $H=\{h\in G/ha_0=a_0\}$, the isotropy group of a_0, is called the unbroken group. Then ϕ can be regarded as a mapping $\phi=S_\infty^{d-1}\to G/H$.

From this and relation (3.8) the mapping ϕ can be identified with the mapping $\mathrm{Pr}\cdot\tilde{A}$ where Pr is the projection

$$\mathrm{Pr}: G\to G/H.$$

The element of $\Pi_{d-1}(G)$ represented by ϕ is then $\mathrm{Pr}*[\tilde{A}]$ where

$$\mathrm{Pr}*: \Pi_{d-1}(G)\to \Pi_{d-1}(G/H) \tag{3.9}$$

is the mapping induced by the projection Pr and $[\tilde{A}]$ is the class of $\tilde{A}$ in $\Pi_{d-1}(G)$.

Now, $\Pi_{d-1}(G)$ is an Abelian group, then $[\tilde{A}]$ is represented by a k-uple of integers $(n_1,\ldots,n_k)$ (the signs and ordering depending on the choice of the generators; note that if $\Pi_{d-1}(G)$ is not free, some of these integers n_i are elements of Z_{p_i}). Then, the equivalence between topological charges of Higgs and gauge fields can be characterized in the following way: *One can choose the generators of $\Pi_{d-1}(G/H)$ in order to have the class of ϕ represented by the same k-uple $(n_1,\ldots,n_k)$ if and only if $Pr*$ is an isomorphism.*

We wish to stress at this point that the interpretation of Yang—Mills fields in terms of connection in the principal fibre bundle and the Higgs field as a global cross section in the associated fibre bundle leads naturally to a condition for the equivalence between Higgs fields and gauge fields topological charges. [in the Abelian case (with $d=2$) studied in Sec. II, $H=\{I\}$, and the precedent discussion was not necessary.] From the exact sequence of homotopy

$$\cdots \Pi_{d-1}(H) \xrightarrow{i*} \Pi_{d-1}(G) \xrightarrow{\mathrm{Pr}*} \Pi_{d-1}(G/H)\to \Pi_{d-2}(H)\to\cdots$$

it can be deduced that Pr* is an isomorphism if

$$\Pi_{d-1}(H)=0, \tag{3.10a}$$

$$\Pi_{d-2}(H)=0. \tag{3.10b}$$

For the case $d=4$, Eq. (3.10b) is always verified since H is compact (H is closed in G), and Eq. (3.10a) reduces to the condition derived by Woo[2]: $\Pi_3(H)=0$. This is the case when the gauge group is broken (i) entirely, (ii) down to a subgroup of Abelian factors. Then, the topological charges of the gauge and Higgs fields are the same.

We consider lastly the example of the group $G=\mathrm{SO}(3)$ and a scalar field forming an isovector $\boldsymbol{\phi}$, with the usual potential $\mathcal{U}(\phi)=\frac{1}{2}\lambda(|\boldsymbol{\phi}|^2-1)^2$, that is, the Gerogi–Glashow model without fermions. The group G can be identified in this case with P^3 (the three-dimensional projective space) and $H=\mathcal{U}(1)$.

If $d=3$, then $\Pi_2(G)=0$. On the other hand, the Higgs field topological charge is characterized by $\Pi_2(G/H)=Z$. What happens in this example is that, although condition (3.10a) is satisfied, (3.10b) is not, since $\Pi_1(H)=Z$. Then, our analysis does not apply. If $d=4$, then $\Pi_3(G)=Z$, and $\Pi_3(G/H)=Z$, that is, a single integer labels the Higgs field topological charge. Note that $A: S_\infty^3\to P^3$ while $\phi: S_\infty^3\to G/H\equiv S^2$.

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